

001443: Code Blue



6. Communicating for safety



8. Recognising and Responding to Acute Deterioration

FOR CODE BLUE – COVID-19 PATIENTS REFER TO PROC:005426 CODE BLUE COVID 19

Background

This procedure is based upon the principles outlined with the National Safety and Quality Health Service Standards (NSQHS) and the Australian Resuscitation Council Guidelines and is endorsed by the Royal Brisbane and Women's Hospital (RBWH) Recognising and Responding to Acute Deterioration (RRAD) committee.

Purpose and intent

This procedure outlines the systems and processes that support the response and management of a Code Blue at the RBWH Herston campus. The RBWH Code Blue system is designed to facilitate timely, reliable and effective detection and response to any person who is experiencing a medical emergency on the RBWH Herston Campus.

Scope and target audience

This procedure applies to all RBWH staff (permanent, temporary, casual and agency) working within RBWH Herston Campus, students on placement within the RBWH, Australian Defence Force (ADF) personnel and volunteers working at the RBWH.

Supporting principles

- All procedures are to be explained to the patient and family with consent obtained in accordance with [001744: Patient consent](#).
- Clinical hand washing is to be done in accordance with [001947: Hand hygiene](#) and is undertaken as required throughout the procedure.
- Invasive clinical procedures are to be undertaken in accordance with [001879: Aseptic Non- Touch Technique \(ANTT\)](#).
- Clinical equipment and environmental surfaces are to be cleaned / decontaminated prior to and after use in accordance with [003139: Cleaning and decontamination-Patient environment and clinical equipment](#).
- The escalation of medical issues for mental health inpatients is to be in accordance with [004617: Escalation of medical issues for mental health inpatients](#).

- Clinical handover is to be undertaken in accordance with [004889: Communicating for Safety](#) whenever there is a change of responsibility and accountability.
- All actions and procedures undertaken are to be documented in accordance with [001072: Documentation in the Patient Record](#).
- Clinical incidents are to be managed in accordance with MNHHS procedure [002044: Clinical Incident and Disclosure Management](#).
- Independent escalation of concerns when the patient or family believe that the patient is deteriorating are undertaken in accordance with [001909: Ryan's Rule](#).

Code Blue overview

A code blue is activated for any patient, member of the public or staff member requiring immediate escalation of care following acute deterioration.

The exception to this is the patient who has limitations of medical care in place, in the form of modifications to medical emergency call criteria (MECC), an Acute Resuscitation Plan (ARP) or an Advanced Health Directive (AHD). If limitations or modifications are in place, then patient escalation is to be carried out in accordance to the respective documentation. Refer [001835: Clinical Observations-Adults](#) procedure.

The RBWH Herston campus is divided into distinct areas for Code Blue response teams. Dependent upon the type and location of the emergency, RBWH Switchboard Services will activate one or multiple teams to provide a Code Blue response. Refer to the [Medical emergency response—Herston campus buildings map](#) and the [Code Blue intranet page](#) for further details. For locations that are beyond the immediate outside surrounds of the RBWH clinical buildings Switchboard Services will connect the call to Queensland Ambulance Service (QAS).

The relevant Code Blue response team will be notified by a paging system from Switchboard Services (except for Code Blue – external medical emergency and neonatal emergency) identifying the type of Code Blue and the location of the emergency. Code Blue team member responsibilities include:

- Code Blue pagers are to be carried by the delegated staff members at all times
- handover of pager at change of shift.

The first and subsequent responders on the scene of a Code Blue initiate first response management including:

- commencing basic life support (BLS) as per the [treatment algorithm](#)
- calling out for help and or pressing the local emergency buzzer. Note: the activation of the red or blue emergency buzzer in clinical areas does not activate the Code Blue Response Team at the RBWH
- facilitating ward resuscitation trolley is brought to the location as able
- ensuring hospital wide help has been activated by calling a Code Blue according to the specific criteria outlined below
- continuous patient assessment and care until the Code Blue team arrives
- recording the event on the [Medical Emergency /& Cardiac Arrest Observation Record form](#)
- providing handover to the Code Blue team.

Code Blue conduct

The following principles are expected of all team members when attending a Code Blue activation.

Team members:

- introduce themselves as they enter the patient's room or the Code Blue location.
- treat each other with respect whilst attending emergency calls.
- operate under the assumption that the Code Blue has been activated based upon genuine clinical concerns and there should be respectful collaboration between the local unit staff and Code Blue team members for ongoing care of the patient.

If conduct of team members is not satisfactory this can be followed up with your line manager or the nurse educator medical emergency response (NE MER).

High risk environments

In all high-risk environments, the responding code blue team will follow the advice of departmental staff regarding the safe environmental management of the patient.

Hyperbaric Services

The Hyperbaric Unit is a high risk; oxygen rich environment and no defibrillation can occur in or near the chamber.

MRI

Patients undergoing MRI examinations will need to be removed from the suite before defibrillation /advanced life support can be initiated. The responding medical emergency team must be cleared by MRI staff prior to entering the MRI suite.

Hydrotherapy pool

Once the patient is out of the pool & in a dry location, a defibrillator can be applied to dry skin. Do not interrupt CPR to attempt to expel or drain clear water or frothy fluid that may re-accumulate in the upper airway during resuscitation.

Types of Code Blue

There are eight different types of Code Blue:

- [Medical Emergency](#)
- [Cardiac Arrest](#)
- [Maternal Cardiac Arrest](#)
- [Category 1 Caesarean Section](#)
- [Obstetric Emergency](#)
- [Neonatal Emergency](#)
- [Paediatric Emergency or Paediatric Cardiac Arrest](#)
- [External Medical Emergency](#).

All staff are to familiarise themselves with the different Code Blue emergency response processes and the emergency equipment available within their work area.

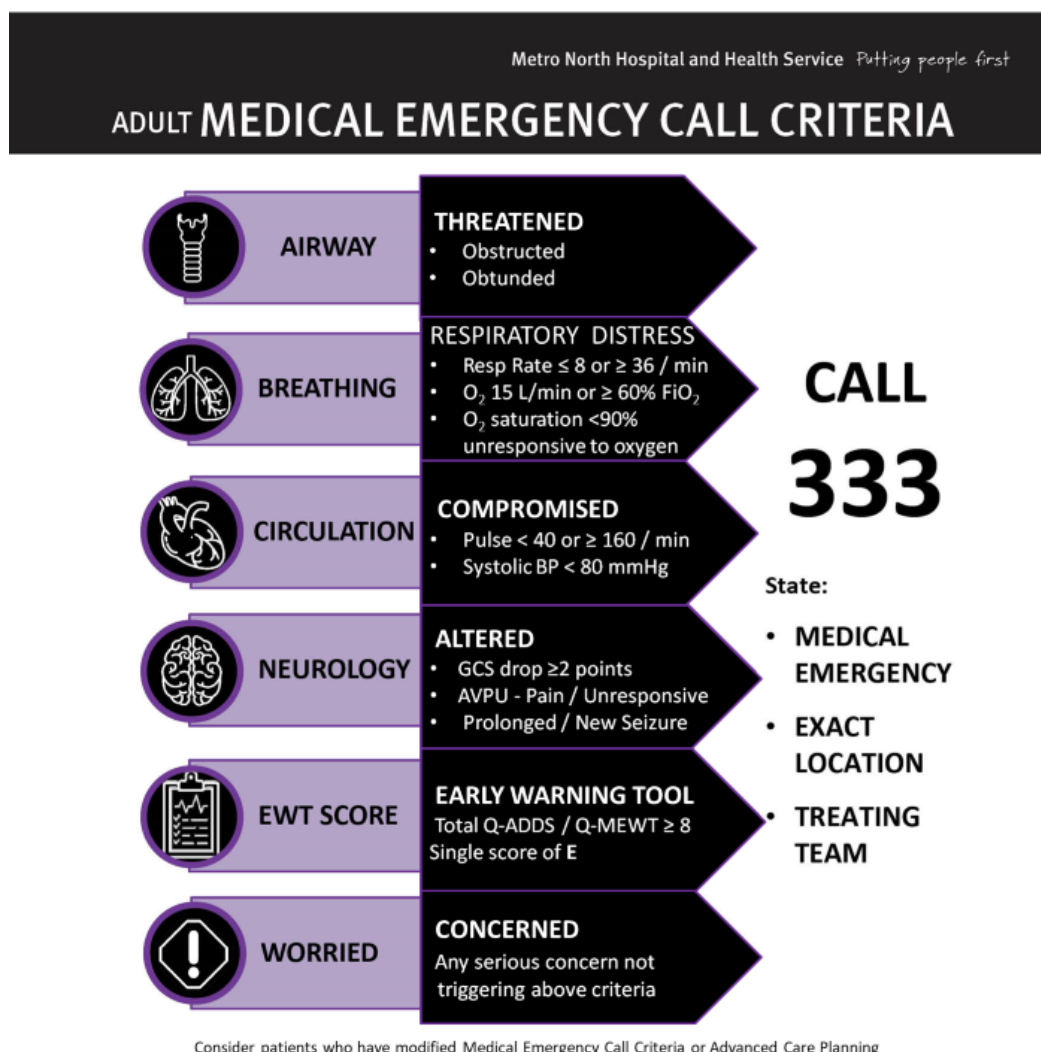
1.0 Medical emergency

Criteria

A Code Blue – medical emergency is to be activated when a patient, member of the public or staff member within the clinical buildings of the RBWH meets the Medical Emergency Call Criteria (MECC). The RBWH clinical buildings include: Joyce Tweddell Building (JTB), Ned Hanlon Building (NHB), James Mayne Building (JMB), Mental Health Centre and Hospital Alcohol Drug Services Units (HADS).

The exception to this is in the case of a patient who has limitations placed in the form of documented modified MECC on the observation chart, an ARP or AHD.

Medical Emergency call criteria



Activation

Any member of staff can activate Code Blue – medical emergency on the RBWH Herston campus by dialling **333** on a hospital phone and stating to Switchboard Services:

- “medical emergency”
- exact location of the emergency e.g. building, floor level, ward, and bed number
- patients treating team if known.

If the patient deteriorates to a cardiac arrest, then the Code-Blue - cardiac arrest process is to be followed.

Medical Emergency Response Team (MERT) attendees and roles

Attendee	Role
Medical registrar (team leader)	Assess the patient situation and develop a plan of action with treating team. Delegate roles, manage transfer of the patient, document the event in the patient record, check and sign documentation of the event on the Medical Emergency & Cardiac Arrest Observation form .
Medical residents	Insert intravenous cannula (IVC), take bloods, gather patient history, assist team leader
Intensive Care Unit (ICU) registrar	Airway management, assist team leader, organise ICU transfer as required.
ICU nurse	Nursing team leader. Bring ICU resuscitation emergency trolley, assists with advanced life support, critical care medications, assists in transport of critically unwell patient to ICU, Operating Theatre (OT), Cardiac Catheter Laboratory, Emergency and Trauma Centre (ETC).
Nurse educator MER (office hours) or UPLATE nurse navigator (afterhours)	Provide clinical support for medical and nursing staff, increasingly active during peri-arrest and arrest situations.
Treating team registrar/consultant	Provide patient information to the team leader. Treating team registrar is to notify the treating team consultant of the Code Blue activation. Initiate end of life care as indicated.

Treating consultant notification

It is mandatory that a treating team registrar attends Code Blue – medical emergencies for their patients. If the registrar is in theatre or off campus, they are still available to discuss the patient's diagnosis and management plan with the Code Blue team.

The treating team consultant is to be notified of **ALL** Code Blue activations for their patients. During office hours the treating team registrar will notify the treating team consultant, afterhours the Code Blue team leader or senior trainee after-hours roster (STAR) registrar will notify the on-call consultant (or delegate).

If there is a decision **NOT** to notify the consultant, it must be fully documented within the patient record.

Specific treatments

Adenosine

The administration of intravenous (IV) adenosine is to occur only within critical care areas of the RBWH or in medical emergency situations where cardiac monitoring (fixed monitoring, telemetry or LIFEPAK 15 or 20) and defibrillator (LIFEPAK 15 or 20) equipment is present and in situ. IV adenosine is to be administered under the guidance of an ICU, cardiology, emergency, medical consultant or senior registrar. Record a continuous Electrocardiograph (ECG) rhythm strip on cardiac monitor or LIFEPAK during administration.

Amiodarone

The administration of IV amiodarone is to occur only within critical care areas of the RBWH or in medical emergency situations where cardiac monitoring (fixed monitoring, telemetry or LIFEPAK 15 or 20) and defibrillator (LIFEPAK 15 or 20) equipment is present and in situ. IV amiodarone is to be administered under the guidance of an ICU, cardiology, emergency, medical consultant or senior registrar. During bolus and infusion administration, perform and record observations every 5 minutes whilst awaiting transfer to critical care areas of the RBWH.

Metoprolol

The administration of IV metoprolol is to occur only within critical care areas of the RBWH or in medical emergency situations where cardiac monitoring (fixed monitoring, telemetry or LIFEPAK 15 or 20) and defibrillator (LIFEPAK 15 or 20) equipment is present and in situ. IV metoprolol is to be administered under the guidance of an ICU, cardiology, emergency, medical consultant or senior registrar. Perform and record observations every minute while injection is in progress. Document reversion and perform Electrocardiogram (ECG).

Digoxin

Refer to [001569: Digoxin Administration](#) procedure.

Referral of mental health inpatient to acute inpatient

Mental health patients who have a medical emergency or cardiac arrest in the mental health centre may require transfer to an acute inpatient ward (e.g. medical or surgical ward). The appropriateness of this is to be discussed between the Code Blue team leader, Mental Health treating team and the relevant on-call consultant for the inpatient team. If the patient requires urgent resuscitation, they are to be transferred to the ICU. In circumstances where there is no ICU bed available, the patient may be transferred to the Emergency and Trauma Centre (ETC) after discussion between the ICU consultant and ETC consultant (ext. 65900).

Transfer to ETC for member of the public and staff members

In the event a Code Blue – medical emergency is activated for a staff member, member of the public or outpatient, the Code Blue response team leader and nominated team members are to escort the patient to the ETC for further assessment and treatment. The ETC triage nurse is to be notified on ext. 67109. If an ETC resuscitation bed may be required, the ETC consultant is to be notified on ext. 65900.

Simultaneous activations

In the event of simultaneous Code Blue activations, the medical registrar (business hours: ext. 70670, all other times: ext. 70489) and ICU registrar (ext. 73596) are to communicate and develop a plan of action to ensure a medical team leader (medical registrar or ICU registrar) is available for each code. A Code Blue-cardiac arrest activation is to take clinical priority.

If an additional Code Blue response team is required:

- the ICU nurse is to liaise with ICU nursing team leader (ext. 65945) to coordinate the formation of the ICU team members
- the medical registrar is to liaise with other medical registrars (e.g. Surgical Treatment and Rehabilitation Service (STAR), Early Patient Intervention Centre (EPIC) registrar) to coordinate the formation of internal medicine team members.

2.0 Cardiac arrest

Criteria

A Code Blue – cardiac arrest is to be activated when a patient, member of the public or staff member within the clinical buildings of the RBWH is:

- unresponsive **AND**
- not breathing normally.

The exception to this is in the case of a patient who has limitations placed in the form of an ARP or AHD.

Activation

Any member of staff can activate Code Blue – cardiac arrest on the RBWH Herston campus by dialling **333** on a hospital phone and stating to Switchboard Services:

- “cardiac arrest”
- exact location of the emergency e.g. building, floor level, ward, and bed number
- patients treating team if known.

Basic life support (BLS)

A patient in cardiac arrest requires immediate BLS to be provided by the first responders. The purpose of BLS is to maintain myocardial and cerebral oxygenation and initiate early defibrillation until advanced life support personnel and equipment are available. [The Australian Resuscitation Council BLS Flowchart](#) provides a structured approach to the assessment and management of a patient requiring resuscitation - **DRSABCD**.

There are three key roles in the provision of BLS:

ROLE 1: FIND THE PATIENT AND PERFORMS DRSABC

- Perform initial assessment of **danger** and patient **response**.
- **Send** for local help by using the local response calling system or shouting loudly.
- Perform an **assessment** of the patients' **airway** and **breathing**.
- If unresponsive and not breathing normally, **commence cardiac compressions**.

ROLE 2: OPERATE THE DEFIBRILLATOR

- Retrieve the resuscitation trolley and bring it to the patient's location on hearing the call for help.
- Apply the defibrillation pads to clean, dry skin, ensuring the pad leads are connected to the defibrillator.
- Safely use the defibrillator as per the RBWH Basic Life Support Manual.
- Remain with the machine unless an ALS trained staff member requests to take over.

ROLE 3: CALL FOR HOSPITAL WIDE HELP THEN MANAGE THE AIRWAY

- Respond to the call for help by dialling **333**
- State "type of Code Blue, exact location (ward and bed number) and treating team if known".
- Manage the patients' airway and ventilate with the manual ventilation bag until medical or advanced life support trained staff arrive.

If the third person is not available, then the second person performs roles 2 and 3.

Cardiac arrest response attendees and roles

Attendee	Role
Medical registrar (team leader)	Ensure advanced life support is provided as per the Advanced Life Support (ALS) algorithm, assess and treat reversible cardiac arrest causes. Delegate roles, manage transfer of the patient, document Code Blue in the patient record, ensure documentation of the event on the Medical Emergency & Cardiac Arrest Observation Form is completed and signed.
Medical residents	Insert IVC, take bloods, gather patient history, assist the team leader.
ICU senior registrar	Airway management, assist the team leader, organise ICU transfer as required. Notify the ICU clinical coordinator / duty intensivist.
ICU nurse	Nursing team leader. Bring ICU resuscitation emergency trolley, assist with advanced life support, critical care medications, assist in transport of the critically unwell patient to ICU, OT, Cardiac Catheter Laboratory, ETC.

Attendee	Role
Nurse educator MER or Up Late nurse navigator	Provide clinical support for medical and nursing staff, facilitate debrief with nursing and medical staff.
Treating team registrar/ consultant	Provide patient information to the team leader. Treating team registrar is to notify the treating team consultant of the Code Blue activation.

3.0 Maternal cardiac arrest

Criteria

A Code Blue – maternal cardiac arrest is activated when a patient, member of the public or staff member who is obviously pregnant or known to be greater >20 weeks pregnant is:

- unresponsive and
- not breathing normally.

Activation

Any member of staff can activate Code Blue – maternal cardiac arrest on the RBWH Herston campus by dialling **333** on a hospital phone and stating to Switchboard Services:

- “maternal cardiac arrest”
- exact location of the emergency e.g. building, floor level, ward, and bed number
- patients treating team if known.

Maternal cardiac arrest response attendees and roles

Attendee	Role
Medical registrar (team leader)	Ensure advanced life support is provided as per the ALS algorithm, assess and treat reversible cardiac arrest causes. Delegate roles, manage transfer of patient, in liaison with obstetric registrar, document Code Blue in the patient record, ensure documentation of the event on the Medical Emergency & Cardiac Arrest Observation Form is completed and signed.
Medical residents	Assist the patient to left lateral tilt position, insert IVC, take bloods, gather patient history, assist the team leader.
ICU senior registrar	Airway management until the anaesthetist is present, assist the team leader, organise ICU transfer as required.
ICU nurse	Nursing team leader. Bring resuscitation emergency trolley, assist with advanced life support, medications, assist in transport of the critically unwell patient to ICU, OT, Cardiac Catheter Laboratory, ETC.

Attendee	Role
Nurse educator MER or UPLATE nurse navigator	Provide clinical support for medical and nursing staff, facilitate debrief with nursing and medical staff.
Treating team registrar/ consultant	Provide patient information to the team leader. Treating team registrar is to notify the treating team consultant of the Code Blue activation.
Obstetric registrar	Attend; in liaison with the team leader: identification and management of obstetric causes for arrest; facilitate perimortem caesarean section if necessary.
Gynaecology registrar	Supported management of patient as guided by team leader.
Midwife	Attend; ensure the perimortem caesarean section pack is taken to cardiac arrest. Assist obstetric registrar with perimortem caesarean section. Receive neonate and assist neonatal team with resuscitation as required.
Anaesthetic registrar or consultant	Alerted by Switchboard Services but only attend if able; assists with maternal resuscitation in liaison with team leader with focus on airway management
Neonatal consultant / registrar/ neonatal nurse practitioner (NNP)	Attends with neonatal resuscitation equipment, resuscitates neonate if of viable gestation, liaises with neonatal consultant.
Neonatal nurse	Bring Cat1 box / emergency equipment from ICN. Assess neonate's requirements for resuscitation. Ensure resuscitation occurs in accordance with NeoResus ARC algorithm. Provide airway, venous access & medication support for the MO/NNP. Ensure documentation of events via Neonatal Emergency Record. Notify Grantley Stable Neonatal Unit (GSNU) of admission if necessary.
Level 5 theatre nurse unit manager (NUM)	Alerted by Switchboard Services but do not attend the patient.
Level 5 theatre shift coordinator	Alerted by Switchboard Services but do not attend the patient.
Level 5 theatre support officer	Alerted by Switchboard Services but do not attend the patient.

Attendee	Role
Patient Support shift coordinator	Alerted by Switchboard Services but do not attend the patient.
Women's and Newborn Services (WNS) bed manager or afterhours nurse manager (AHNM)	Alerted by Switchboard Services but do not attend the patient.

4.0 Category 1 caesarean section

Criteria

A Code Blue – Category 1 caesarean section is called when there is an immediate need for a Category 1 caesarean section to preserve the life of the mother, the foetus or both. Indications for Code Blue – Category 1 caesarean section activation include but are not be limited to:

- umbilical cord prolapse
- sustained foetal bradycardia ≤ 80 bpm for > 5 minutes
- severe antepartum haemorrhage (APH) associated with placental abruption or placenta previa
- uterine rupture
- foetal scalp pH < 7.15 / lactate > 5.0 mmol/l
- vasa praevia - antepartum haemorrhage (APH) probably detected as foetal distress
- failed breech extraction for a second twin
- failed vaginal breech delivery
- failed instrumental delivery with foetal compromise.

Activation

Any member of staff can activate Code Blue –Category 1 caesarean section on the RBWH Herston campus by dialling **333** on a hospital phone and stating to Switchboard Services:

- “Category 1 caesarean section”
- exact location of the emergency e.g. building, floor level, ward, and bed number
- patients treating team if known.

Additionally, the Level 5 theatre coordinator is to be contacted via ext. 65969.

Category 1 caesarean section response attendees and roles

Attendee	Role
Obstetric registrar, Birth Suite	Attend and act as team leader Assess the patient and initiate Category 1 caesarean section management plan, ensure the on-call consultant is aware and attend when necessary

Attendee	Role
	Ensure category 1 activation has been initiated and theatre coordinator is notified prior to transfer to theatre.
Gynaecology registrar, Birth Suite	Attend and can act as team leader if obstetric registrar involved in direct patient management. Supported management of patient as guided by the team leader
Anaesthetic registrar, WNS	Attend and assist with resuscitation as needed in particular: airway and circulatory support Preparation for transfer to theatre (if not already in theatre).
Anaesthetic technician	Provide assistance to the anaesthetic registrar.
Birth Suite team leader (or other midwife as delegated by birth suite team leader)	Confirm nature of obstetric emergency and management plan. Delegate clinical tasks to attending medical and midwifery staff including scribe to document events. Assist with clinical tasks as required and ensure emergency is managed appropriately and in a timely manner.
Neonatal consultant / registrar / NNP	Immediately attend, obtain history from obstetric registrar, calls neonatal consultant if required. Resuscitates the neonate.
Neonatal nurse	Bring Cat1 box with emergency equipment from ICN. Assess neonate's requirements for resuscitation. Ensure resuscitation occurs in accordance with NeoResus ARC algorithm. Provide airway, venous access & medication support for the MO/NNP. Ensure documentation of events via Neonatal Emergency Record. Notify GSNU of impending admission as required.
Patient support officer, Birth Suite	Assist with manual handling and patient transfer in theatre.
AHNM	Alerted by Switchboard Services but do not attend the patient.
Afterhours CNC theatres	Alerted by Switchboard Services but do not attend the patient.
Level 5 theatre NUM, shift coordinator and support officer	Alerted by Switchboard Services but do not attend the patient.

Attendee	Role
Patient Support shift coordinator	Alerted by Switchboard Services but do not attend the patient.
WNS bed manager or AHNM	Alerted by Switchboard Services but do not attend the patient.

5.0 Obstetric emergency

Criteria

A Code Blue – obstetric emergency is called when there is a need for immediate medical and or obstetric attention for a maternity patient, other than a Category 1 caesarean section.

Within Birth Suite/Birth Centre an emergency buzzer on the wall is pressed and this initiates an immediate response from obstetric staff on level 5. The Birth Suite team leader, Birth Suite registrar, Birth Suite consultant (if on site) attend at the sound of this alarm. If further assistance is required 333 can be dialled at the discretion of the obstetric team.

Elsewhere on RBWH campus dials **333** on a hospital phone and advise switch of obstetric emergency.

If the neonatal team is required dial **64333** on a hospital phone or press a neonatal emergency buzzer if available (Birth Suite/Birth Centre).

Activation

Any member of staff can activate Code Blue – obstetric emergency on the RBWH Herston campus by dialling **333** on a hospital phone and stating to switch board services:

- “obstetric emergency”
- exact location of the emergency e.g. building, floor level, ward, and bed number
- patients treating team if known.

Obstetric emergency response attendees and roles

Attendee	Role
Obstetric registrar, Birth Suite	Attend and act as team leader. Confirm the nature of the obstetric emergency. Assess the patient and initiate appropriate obstetric emergency management plan. Determine if further assistance is required.
Team leader, Birth Suite	Attend and confirm the nature of the obstetric emergency and management plan. Delegate clinical tasks to attending medical and midwifery staff including scribe to document events. Assist with clinical tasks as required and ensure emergency is managed appropriately and in a timely manner.

Attendee	Role
	Liaise with theatre and neonatal team as necessary.
Gynaecology registrar, Birth Suite	Attend if available. Can act as team leader if obstetric registrar involved in direct patient management. Support management of patient as guided by team leader.
Anaesthetic registrar, WNS	Alerted by Switchboard Services but only attend if able. Assist with resuscitation as needed in particular: airway and circulatory support.
Level 5 theatre NUM and shift coordinator	Alerted by Switchboard Services but do not attend the patient.
Patient support staff.	Alerted by Switchboard Services but do not attend the patient.
WNS bed manager or AHNM	Alerted by Switchboard Services but do not attend the patient.

6.0 Neonatal emergency

Criteria

A neonatal emergency is any event which threatens the life and wellbeing of a neonate. This includes altered physiology, seizures, haemorrhage, altered muscle tone or level of responsiveness.

For boarder infants over 28 days activate a Code Blue – Paediatric Emergency. Refer to [000067: Boarder Infant Management](#).

Activation

Any member of staff can activate Code Blue – neonatal emergency within the Birth Suite or GSNU by pressing the emergency call buzzers.

Any member of staff can activate Code Blue – neonatal emergency within the post-natal, peri-operative suite or ETC by dialling **64333** on a hospital phone and state to the neonatal nurse:

- “neonatal emergency”
- exact location of the emergency e.g. building, floor level, ward, and bed number
- briefly identify the neonate's primary problem.

Neonatal emergency response attendees and roles

Attendee	Role
Neonatal consultant / registrar / NNP	Attend, assess neonate, lead resuscitation, call for back up if needed

Neonatal nurse	Bring emergency equipment from ICN. Assess neonate's requirements for resuscitation. Ensure resuscitation occurs in accordance with NeoResus ARC algorithm. Provide airway, venous access & medication support for the MO/NNP. Ensure documentation of events via Neonatal Emergency Record. Notify GSNU of admission if necessary
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7.0 Paediatric medical emergency or paediatric cardiac arrest

Children may attend the RBWH from other health care facilities (for example, the Queensland Children's Hospital (QCH)) or are referred directly to use RBWH services for treatment.

Criteria

A Code Blue – paediatric emergency is to be activated when a patient / member of the public (aged 14 years and under) or boarder infant meets the Paediatric Emergency Call Criteria unless the criteria has been modified by the treating consultant.

Any observation within the purple area on the Children's Early Warning Tool (CEWT)	Total CEWT score ≥ 8
Airway threat	Apnoea
Sedation score of 3	Seizure
Bleeding	Any concern or worry

Activation

Any member of staff can activate Code Blue – paediatric emergency on the RBWH Herston campus by dialling **333** on a hospital phone and stating to Switchboard Services:

- “paediatric medical emergency / cardiac arrest”
- exact location of the emergency e.g. building, floor level, ward, and bed number
- patients treating team if known.

Switchboard Services will phone the ETC emergency phone 67304 and notify them of the Code Blue – paediatric medical emergency. To notify the response team, the ETC staff activate the local alert system by pressing the emergency buzzer located on the resuscitation side of the ETC work station (*fishbowl*). Switchboard Services must notify the child's treating consultant at the time of alerting the paediatric medical emergency response team.

If escalation of care to a paediatric care facility is required the treating consultant is to notify Retrieval Service Queensland on 1300 799 127. Refer [003962: Clinical Observations and Escalation of Care for Children](#).

Paediatric medical emergency response attendees and roles

Attendee	Role
ETC medical officer	Attend, assess patient situation and develop plan of action with treating team. Delegate roles, manage transfer of patient as required, document Code Blue in patient record.
ETC nurse	Brings paediatric emergency backpack, assists MO with patient assessment, defibrillation and medications, escorts patient as required.
Treating team registrar and consultant	Provide patient information to the team leader.
Nurse educator MER	Provide clinical support for medical and nursing staff, increasingly active during peri-arrest and arrest situations

8.0 External medical emergency

The RBWH external medical emergency response team responds to medical emergencies and/or cardiac arrests in locations within the immediate outside surrounds of the RBWH clinical buildings. The focus of care for these patients is to be stabilisation in preparation for transfer to ETC or home ward, if an inpatient.

Activation

Any member of staff can activate Code Blue – external medical emergency on the RBWH Herston campus by dialling **333** on a hospital phone or 364 61600 from a public phone and stating to Switchboard Services:

- “external medical emergency”
- type of the emergency e.g. adult or paediatric
- exact location of the emergency e.g. building, floor level
- patient’s treating team if known.

Switchboard Services will phone the ETC emergency phone ext. 67304 to advise of the Code Blue – external emergency. To notify the external response team, the ETC staff activate the local alert system by pressing the emergency buzzer located on the resuscitation side of the ETC work station (*fishbowl*).

Alternatively, a member of the public may approach the ETC triage desk for external emergency assistance. To notify the external response team, the ETC staff activate the local alert system by pressing the staff assist buzzer at the triage desk and the External Emergency buzzer located on the resuscitation side of the ETC work station (*fishbowl*).

For some locations on the RBWH Herston campus Switchboard Services will connect the caller to Queensland Ambulance Service (QAS).

External emergency response attendees and roles

Attendee	Role
ETC medical officer	Assess patient situation, stabilise and develop plan of action.
ETC nurse	Assists medical officers, brings emergency backpack and defibrillator
ETC patient support officer	Provides patient trolley/ wheelchair / back board and assists with manual handling as required. Assist with transfer of patient to ETC or home ward.
Treating team registrar/ consultant (if inpatient)	Provide patient information to the team leader.
Nurse educator MER	Provide clinical support for medical and nursing staff, increasingly active during peri-arrest and arrest situations.
Protective Services	Protects people, places and property from potential threats at external MER location.

Code Blue team stand-down

The Code Blue Team leader is able to stand down the Code Blue team once one or more of the following criteria are met:

- patient is stable (e.g. physiological parameters are no longer within MECC or within modified parameters established by the Code Blue team leader) and the treating team has assumed responsibility to manage the patient on the ward
- patient has been transferred to a higher acuity area
- decision to set in place limitations of care and manage on the ward
- patient is deceased.

Post Code Blue management

The patient is to be identified in the Patient Flow Manager as a "Patient of Concern".

Documentation

The [Medical Emergency and Cardiac Arrest Observation Record Form](#) is to be completed during a Code Blue event and this role will be allocated by the team leader to a staff member who is present at the event. The completed [Medical Emergency & Cardiac Arrest Observation Form](#) is to be signed by the medical team leader at the completion of the Code Blue and is to be filed within the patient clinical record.

The ward nurse is to transcribe the last set of vital signs from the [Medical Emergency & Cardiac Arrest Observation Form](#) to the patient's observation form to assist with tracking and trending.

All administered medications documented upon the medical emergency and cardiac arrest observation record must be transcribed onto a Medication Chart by the prescribing medical officer as soon as possible.

The attending team leader is to document assessment details, interventions, investigation performed, outcomes and plan for ongoing care in the patient's medical record.

If a decision is made not to resuscitate or to limit resuscitation attempts, an ARP should be implemented with appropriate consultation and documentation. Once decisions relating to resuscitation and ceilings of treatment have been made:

- The ARP must be completed and placed at the FRONT of the Patient Record
- The purple SW065a Resuscitation Plan Exists label MUST be affixed to the Batch Cover Sheet, which is in the plastic sleeve at the front of the Current Encounter Chart (CEC).

Debriefing

Opportunities are to be created for staff involved in clinical emergencies to debrief following the event. Debriefing can be initiated by the staff present at the event and can be either formal or informal. The debriefing process should be a safe platform for staff to express their reactions, reflect on their practice while providing opportunity for professional development. If further debriefing is required on a group or individual basis, this should be discussed with the nurse unit manager or clinical director.

Equipment

Standard resuscitation trolley

Standardised resuscitation trolleys are located within clinical areas across the RBWH. The equipment and medications contained upon the resuscitation trolleys has been endorsed by the RBWH RRAD committee to facilitate the initial management of a cardiac arrest. No changes are to be made to equipment or supplies on the resuscitation trolley without the authority of the nurse educator MER.

All trolleys are to be checked at the start of each shift and after every use, using the [Standard resuscitation trolley equipment list](#) and [checking guidelines](#) located on the [RBWH Code Blue Intranet site](#) to confirm the functioning and availability of all equipment. It is the responsibility of the shift co-ordinator to ensure the resuscitation trolley is checked and the check list is signed.

Any issues identified during the checking process are to be immediately actioned by the staff members undertaking the check. Actions are to be noted on the checklist and followed up as necessary with the team leader. Stock can be replaced by following the instruction on the Resuscitation trolley check list. Any additional equipment is to be removed or discussed with the nurse educator MER.

Intensive Care Unit (ICU) trolley

The ICU emergency response trolley has an expanded list of equipment and medications potentially required during the provision of ALS. The ICU trolley has a Lifepak 15 defibrillator capable of providing haemodynamic monitoring, end-tidal capnography, cardiac monitoring, pacing and manual defibrillation.

Defibrillators

At the RBWH there are four types of Lifepak defibrillators including the Lifepak 1000, 15, 20 and CR2. Defibrillators are placed in key locations throughout the RBWH Heston Campus and upon resuscitation trolleys. It is the responsibility of nursing, medical and allied health staff to be familiar with the operation of the defibrillator within their clinical area.

Biomedical Technology Services (BTS) test defibrillators each year. Any problems or faults identified by clinical staff are to be immediately reported to BTS. Loan defibrillators are available from Central Equipment Loans Service (CELS).

Paediatric equipment

Paediatric basic life support equipment must be available in areas providing services to children aged 14 years and under. Paediatric equipment can be sourced from CELS when a child aged 14 years and under is attending for care. The list of [paediatric emergency equipment](#) is located on the RBWH Code Blue intranet.

Code Blue training and education

Basic life support

Medical officer

Induction

Initial BLS assessment at Medical Officer Orientation Program (or within 30 days of commencement).

Ongoing

Annual training and assessment conducted by Nurse Educator MER or CPR Resource Person. Medical Officer BLS sessions and booking information can be found at:
<http://hi.bns.health.qld.gov.au/education/basic-life-support.htm>

Nursing

All nurses, midwives and assistants in nursing (AINs) are mandated to be assessed as competent in BLS.

Induction

Initial BLS assessment at nursing & midwifery service / service line orientation program (or within 30 days of commencement).

Ongoing

Annual training and assessment conducted by nurse/midwifery educators or resource person.

Allied health

All clinical allied health staff that are involved in direct patient care are mandated to be assessed as competent in BLS.

Induction

Initial BLS training and assessment conducted by resource person within three months of commencement.

Ongoing

Annual BLS & AED training and assessment conducted by resource person annually.

BLS resource persons

At least one member of staff from each clinical area is to be trained as a cardiopulmonary resuscitation (CPR) resource person. The expectations and responsibilities of the CPR Resource Person are outlined in the CPR Resource Person Manual.

Intermediate life support

Nursing staff in areas where procedural sedation is administered must complete ILS training and assessment as nominated by work unit requirements. The eight-hour RBWH Nursing ILS course includes a written assessment and a practical clinical skill assessment as per the '*Adult Intermediate Life Support Practical Assessment Tool*' conducted by ILS course coordinator and ILS Resource Person. To demonstrate competency, nursing staff are to complete a written and practical assessment yearly.

Advanced life support

Medical officer

Medical officers rostered as the team leader of a Code Blue – Medical Emergency or Cardiac Arrest Team must complete ALS training. The four-hour RBWH Adult MERT ALS course includes a written assessment and a practical clinical skill assessment as per the '*Adult Advanced Life Support Annual Assessment Tool*' conducted by an ALS Resource Person. Medical Officers are to attend at a minimum every second year.

Recognition of prior learning

Recognition of prior learning for ALS competency is available for clinical staff following application to the Nurse Educator MER.

Nursing

Nursing staff in critical care areas must complete ALS training and assessment as nominated by work unit requirements. These areas are the Coronary Care Unit, Emergency and Trauma Centre and Intensive Care Unit. The eight-hour RBWH Nursing ALS course includes a written assessment and a follow up practical clinical skill assessment as per the '*Adult Advanced Life Support Annual Assessment Tool*' conducted by an ALS Resource Person. To demonstrate competency, nursing staff are to complete a written and practical assessment yearly.

Women's and Newborns

All clinical staff working in midwifery and neonatology must also complete annual neonatal resuscitation competency assessment.

Multidisciplinary obstetric emergency training is undertaken for all Women's & Newborn Services (WNS) maternity staff via the Practical Obstetric Multi-Professional Training (PROMPT) skills workshop held monthly with the expectation that the consultants attend every two years.

First Aid

Non-clinical work areas of the RBWH are required to undertake a first aid risk assessment to determine the level of risk associated with first aid requirements in the area. Work areas identified as high risk and above or as required through legislation need to maintain first aid kits and trained first aid personnel Metro North Hospital and Health Service (MNHHS) procedure [003383 First Aid](#).

Electrical workers and first aid officers must undertake BLS training on commencement of employment and complete an annual BLS competency assessment in accordance with legislative requirements (Electrical Safety Regulation 2002 and Work Health and Safety Act 2011) and procedure 87201/Proc: First Aid – Provision of).

Childcare staff at Lady Ramsay Early Learning and Education Centre are required to hold a current First Aid, CPR, Anaphylaxis Emergency Management and Asthma Emergency Management Certificates in accordance with the Education and Care Services National Queensland Act (2011) and Education and Care Services National Regulations (2011).

Partnering with consumers

Patients and family members are to be encouraged and given the opportunity to ask questions, clarify information and identify goals of care during communication processes. Staff are responsible for providing information in a way that is understandable and meets patients' needs and are to use perception checking techniques to ensure patient understanding of discussions.

Legislation and other authority

Electrical Safety Regulation 2002 and Work Health and Safety Act 2011

Education and Care Services National Queensland Act (2011) and Education and Care Services National Regulations (2011)

Human Rights Act 2019 (Qld)

References

Australian Commission on Safety and Quality in Health Care (2010). National Consensus Statement: Essential Elements for Recognising and Responding to Clinical Deterioration.

Australian Resuscitation Council (2011) Guideline 11.10 – Resuscitation in special circumstances. Available from <http://resus.org.au/guidelines/>

Jeejeebhoy FM, Zelop CM, Windrim R, Carvalho JC, Dorian P, Morrison LJ. Management of cardiac arrest in pregnancy: A systematic review. *Resuscitation*. 2011 Jul;82(7):801-9.

Jeejeebhoy FM, Windrim R. (2014) Management of cardiac arrest in pregnancy. *Best Practice & Research Clinical Obstetrics and Gynaecology* 28 (2014): 607-618.

National Safety and Quality Health Service Standards 2nd edition

Queensland Government (2014) [First aid in the workplace Code of practice 2014](#)

Related documents

- Palliative and Supportive Care Service (PSCS) is available to all patients at the RBWH with potentially life-limiting illness regardless of diagnosis. Referrals can be made in accordance with the following documents:
 - [Palliative Care Service referral form \(MR 6049\)](#)
 - [Palliative and Supportive Care Service Referral Guidelines](#)
 - [Clinical Guidance for the Dying Patient - Symptom Management Guidelines.](#)
- [Standard resuscitation trolley equipment list](#)
- [Standard resuscitation trolley checking guidelines](#)
- [Standard resuscitation trolley – daily check record](#)
- [Paediatric emergency equipment list](#)
- MNHHS procedure [003383: First Aid](#)

Appendix 1- Definition of terms

Term	Definition
Neonate	Newborn infant within first 28 days of life
Paediatric patient / child	28 days – 14years of age and under

Document history

Author	Nurse Educator - Medical Emergency Response
Custodian	Chairperson Recognising and Responding to Acute Deterioration Committee
Compliance evaluation and audit	Medical emergency and cardiac arrest data is entered into the SQU database with activity and trends reviewed by RRAD committee, with escalation and action as indicated. Issues and incidents reported via organisational clinical incident system (RiskMan) with escalation to local clinical incident review committee and Management Advisory Committee and further action undertaken as indicated.
Replaces Document/s	005200: Code Blue – Q-ADDS Trial Only V1 001443: Code Blue V4
Consultation	Members, Recognition and response to acute deterioration committee RBWH executive directors, nursing/midwifery directors, assistant nursing directors for onwards dissemination to clinical teams Facility service director, for further dissemination to relevant stakeholders BBWH safety and quality officers Quality Innovation and Patient Safety Service
Marketing Strategy	RBWH Intranet PPP site “What’s New Site” and local dissemination
Key words	001443; code blue; medical emergency; cardiac arrest; emergency response; emergency response team; MERT; cardiopulmonary resuscitation; CPR; obstetric emergency; CAT1 caesarean section; perimortem caesarean section; maternal cardiac arrest; MECC

Custodian Signature

Date

**Chairperson Recognising and Responding to Acute Deterioration Committee
Royal Brisbane and Women's Hospital**

AUTHORISATION

Authorising Officer Signature

Date

Executive Director Medical Services, Royal Brisbane and Women's Hospital

**Incident Management Team Commander, Emergency Operation Centre
Royal Brisbane and Women's Hospital**

Date

The original signed version is retained by the RBWH Quality Innovation and Patient Safety Service (QIPSS)